

P1 — Analytic Cohort (Lab 2)

Model submission — demonstration paper (Kueh et al., BMJ Open 2026).

Study question

Among U.S. adults meeting FLI-based MASLD criteria, is central adiposity (WHtR) associated with all-cause mortality, and how robust is that association to how the analytic cohort is constructed?

Scientific (locked) estimand — IV vs III. Among adults meeting FLI-based MASLD criteria, do **non-obese adults with high central adiposity** have higher all-cause mortality than **non-obese adults with low central adiposity**?

Replication estimand — IV vs I. The paper references its contrasts to Group I (obese, low central adiposity). We reproduce that, because reproducing it is what *Replicate* means. Every increment and milestone states which of the two it is reporting.

Source files and merge key

We assembled six NHANES cycles (2007–2008 through 2017–2018) from the public component files: demographics (DEMO), body measures (BMX), fasting biochemistry for triglycerides and glucose (TRIGLY/GLU), standard biochemistry for GGT and creatinine (BIOPRO), blood pressure (BPX), glycohaemoglobin (GHB), lipids (HDL/LDL/TCHOL), smoking (SMQ), alcohol (ALQ), physical activity (PAQ), and the medical-conditions and prescription-medication questionnaires (MCQ, RXQ). These were merged to the NCHS 2019 public-use linked mortality file. **All merges key on the respondent identifier SEQN**, one row per person per cycle, cycles stacked before selection. FLI is computed as $FLI = 100 / (1 + \exp(-(0.953 \cdot \ln(TG) + 0.139 \cdot BMI + 0.718 \cdot \ln(GGT) + 0.053 \cdot waist - 15.745)))$.

Eligibility decisions

1. **Adults only** — restrict to age ≥ 18 (RIDAGEYR). Unambiguous.
2. **Fasting subsample membership (DESIGN)** — fasting analytes are measured in a *designed* NHANES subsample, and WTSAF2YR exists to account for that selection stage and its nonresponse. This is not an eligibility choice we make, and it is not missing data.
3. **FLI computable (MISSINGNESS)** — among those already *in* that subsample, some still lack triglycerides, GGT, BMI or waist. That is item nonresponse: a different problem from step 2, and the only one of the two that a missing-data method could address.
4. **Steatosis** — $FLI \geq 60$, the paper’s stated cut.
5. **MASLD** — steatosis plus ≥ 1 of five cardiometabolic criteria (elevated BMI/waist, hyperglycaemia, hypertension, high triglycerides, low HDL). We coded the standard 2023 nomenclature set; only 2 people were lost here, so criterion wording is not load-bearing.
6. **Mortality-eligible** — retain records with a valid NCHS mortality linkage eligibility flag and non-missing follow-up time (censored 31 Dec 2019).
7. **Phenotype grouping** — the four groups partition on obesity and central adiposity (see BMI note below).

Cohort funnel

Every step carries a type label. A funnel shows N falling; it does not show *why*, and the reasons are not interchangeable.

Step	Criterion	N	Type	Dropped
0	NHANES 2007–2018 records	59,842	—	—

Step	Criterion	N	Type	Dropped
1	Age ≥ 18	36,580	ELIGIBILITY	23,262
2	In fasting subsample	14,962	DESIGN	21,618
3	FLI computable (TG + GGT + BMI + waist)	14,170	MISSINGNESS	792
4	Steatosis (FLI ≥ 60)	6,052	TARGET POPULATION	8,118
5	MASLD (≥ 1 cardiometabolic criterion)	6,050	TARGET POPULATION	2
6	Phenotype group assignable	6,050	MISSINGNESS	0
7	Mortality-eligible and linked	6,048	OUTCOME ASCERTAINMENT	2

After eligibility, the two largest drops are **Step 2 (design)** and **Step 4 (target population)**, and *neither is missing data*.

- Step 2 is the NHANES fasting subsample. It was **selected deliberately**, and WTSF2YR exists to account for that selection stage and its nonresponse. The correct response is the right weight and a domain analysis (P4), not imputation.
- Step 4 is people without hepatic steatosis. They are not missing — they are **not who the paper is about**.
- **Item nonresponse: 792 people** — selected into the fasting subsample, still without a computable FLI. **Outcome ascertainment: 2 people** unlinked to mortality follow-up. These are different problems and we do not add them into a single “missing” total.

We separate these because collapsing them is the standard error here. Reading the funnel as “41% of adults have a computable FLI, so 59% are missing” merges a designed subsample with a disease definition and produces an apparent missing-data problem of about 29,700 people where the real one is $792 + 2$. The 792 are a real item-nonresponse problem and they remain **an upstream issue we do not resolve**: they sit before the analytic cohort, so no within-cohort method reaches them. P5 interrogates a different and smaller thing — the **109 missing cancer values** among the Model-1 covariates *inside* the cohort. Both are genuine; only the second is tractable here.

Note also that our cohort is **6,048**, not the 6,371 the pipeline reports for the full analytic file: restricting to records with a valid fasting weight is what makes the design-aware analysis possible, and it is a different (correct) denominator.

Reproduced group sizes

Group (BMI / WHtR)	Ours	Paper
I — obese / low central (WHtR<0.6)	391	386
II — obese / high central (WHtR $\geq$ 0.6)	4,585	4,541
III — non-obese / low central	787	769
IV — non-obese / high central	608	604

All four groups reproduce within about 2% (the largest gap is Group III, 787 against 769). Group IV (non-obese, high central adiposity) is the paper’s headline “worst survival” phenotype; our all-cause death counts

(5 / 389 / 74 / 118; total 586) likewise match the paper’s (5 / 388 / 74 / 118; total 585).

Documented ambiguity: alcohol operationalization

The paper’s Methods state an exclusion of “>2 drinks/day (men), >1 (women),” but its reported N and deaths match **no effective exclusion**. We tested three defensible codings of the same sentence:

- **Crude ALQ130 rule** (usual drinks on drinking days, applied flatly): removes **2,063** $\rightarrow$ N = 4,308.
- **Frequency-weighted average** (drinks/day = quantity $\times$ frequency $\div$ reference period): removes **427** $\rightarrow$ N = 5,944.
- **No effective exclusion** (as the published counts imply): removes **0** $\rightarrow$ N = 6,371.

One coding trap, worth its own line. ALQ130 uses NHANES reserved codes — **777** (“refused”) and **999** (“don’t know”). Applied to raw values, a rule like **ALQ130 > 2** silently reclassifies every refusal as a heavy drinker; here that is nine people, and it moves the crude-rule exclusion from 2,063 to 2,072. Reserved codes are not measurements. Strip them before any threshold rule, in every increment.

The three codings span a >2,000-person range from one ambiguous sentence. We **did not tune the coding to hit the paper’s N**. We adopted the no-exclusion path because it reproduces the published cohort and is the most transparent about the discrepancy, and we carry the alcohol question forward as a documented methods-vs-results inconsistency rather than a solved step.

BMI cut-point finding

The paper’s Table headers label the non-obese groups “BMI < 25.” That is inconsistent with the data: the non-obese groups have BMI **medians of 27.6 (III) and 28.8 (IV)** — impossible under a 25 cut. The Methods text uses BMI ≥ 30 for obesity, and **coding the split at 30 reproduces the group sizes above**. We conclude the “< 25” headers are inconsistent with the analysis as run, and that the operative threshold is 30.

Honest account of the residual N gap

We reproduce 6,371 on the full analytic file where the paper reports **6,300**, and lock **6,048** as the analysis domain (those with a valid fasting weight). We have not fully explained the remaining difference of 71: both alcohol exclusions above land further from the paper’s figure than no exclusion does, so we report the gap as unresolved rather than attribute it. Completeness on the paper’s Model-1 covariates is reported in P5, which is where it is analysed; an earlier draft quoted a 6,246 figure from an expanded covariate set that Model 1 does not contain, and that figure is withdrawn.

Reproducibility notes

Analysis in R; cycles stacked and merged on **SEQN** in a single deterministic script. FLI coefficients are hard-coded from the source formula. Every funnel row is emitted by the script so counts are re-derivable end to end, and the three alcohol codings toggle from one parameter. Group definitions use BMI ≥ 30 and WHtR ≥ 0.60 .